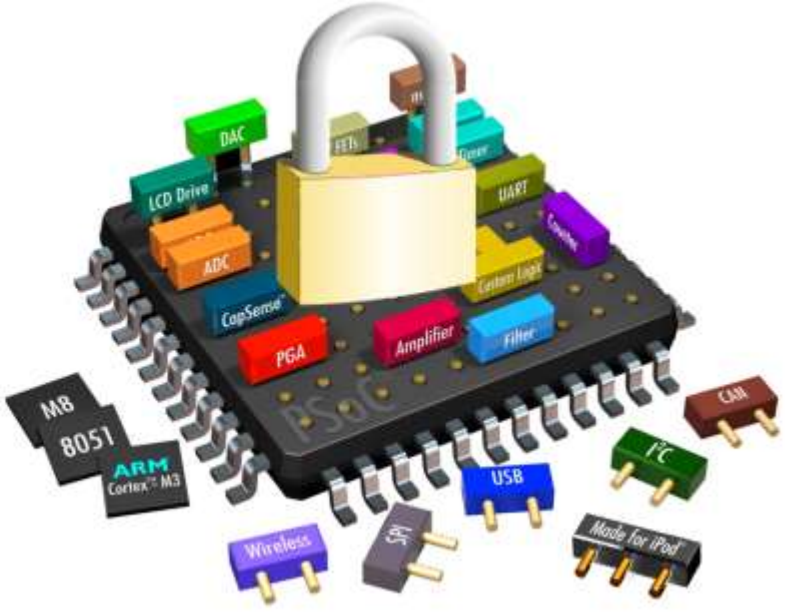
Hardware Security

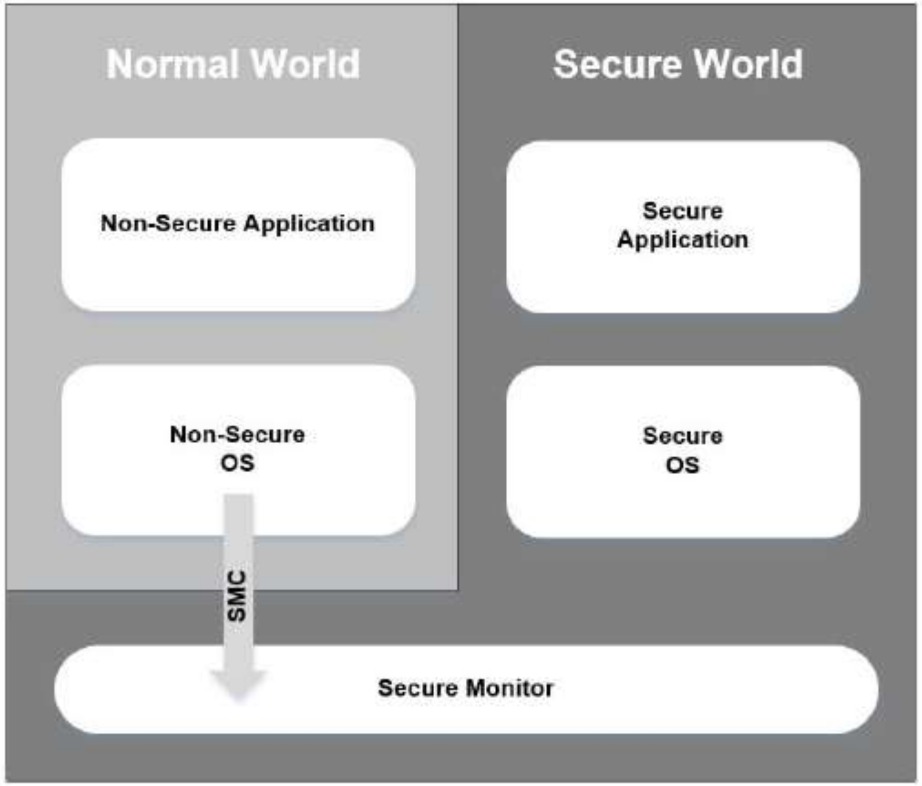
TEE Examples: ARM TrustZone



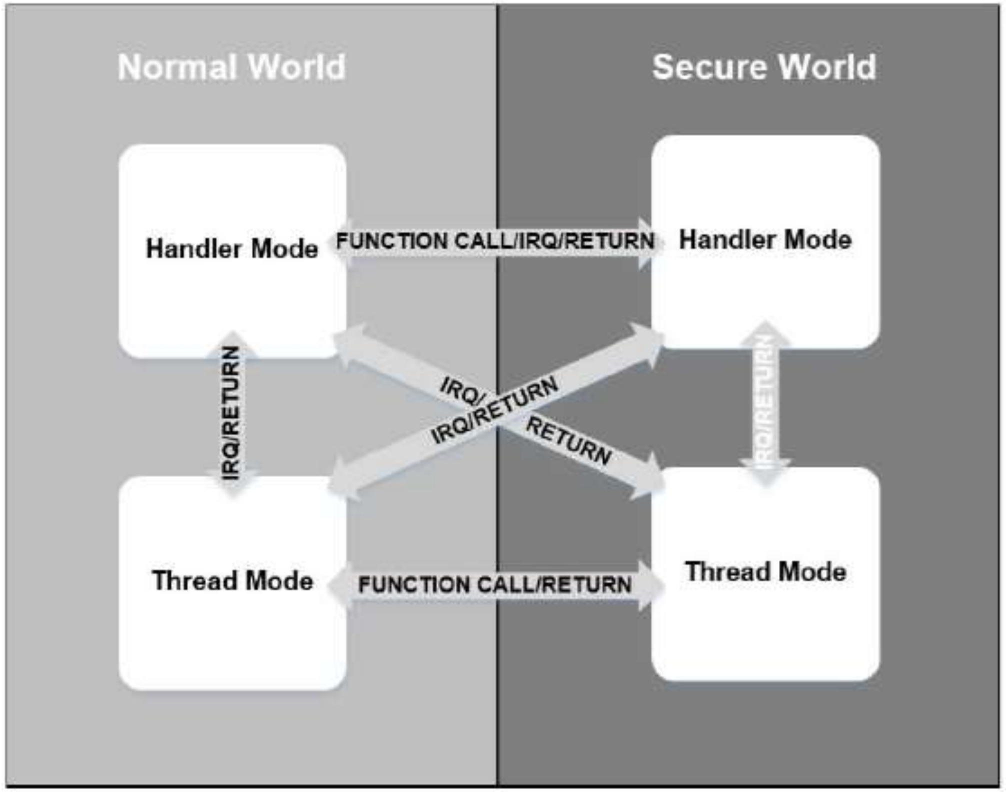
Security States

- With TrustZone Security Extensions, processors have two security states
- Secure State (Secure World)
  - Everything that runs when processor state is secure. Has access to all resources including secure memory regions.
- Non-Secure State (Normal World)
  - Everything that runs when processor is in non-secure state. Hardware barriers prevent access to secure world resources.

Software Architecture

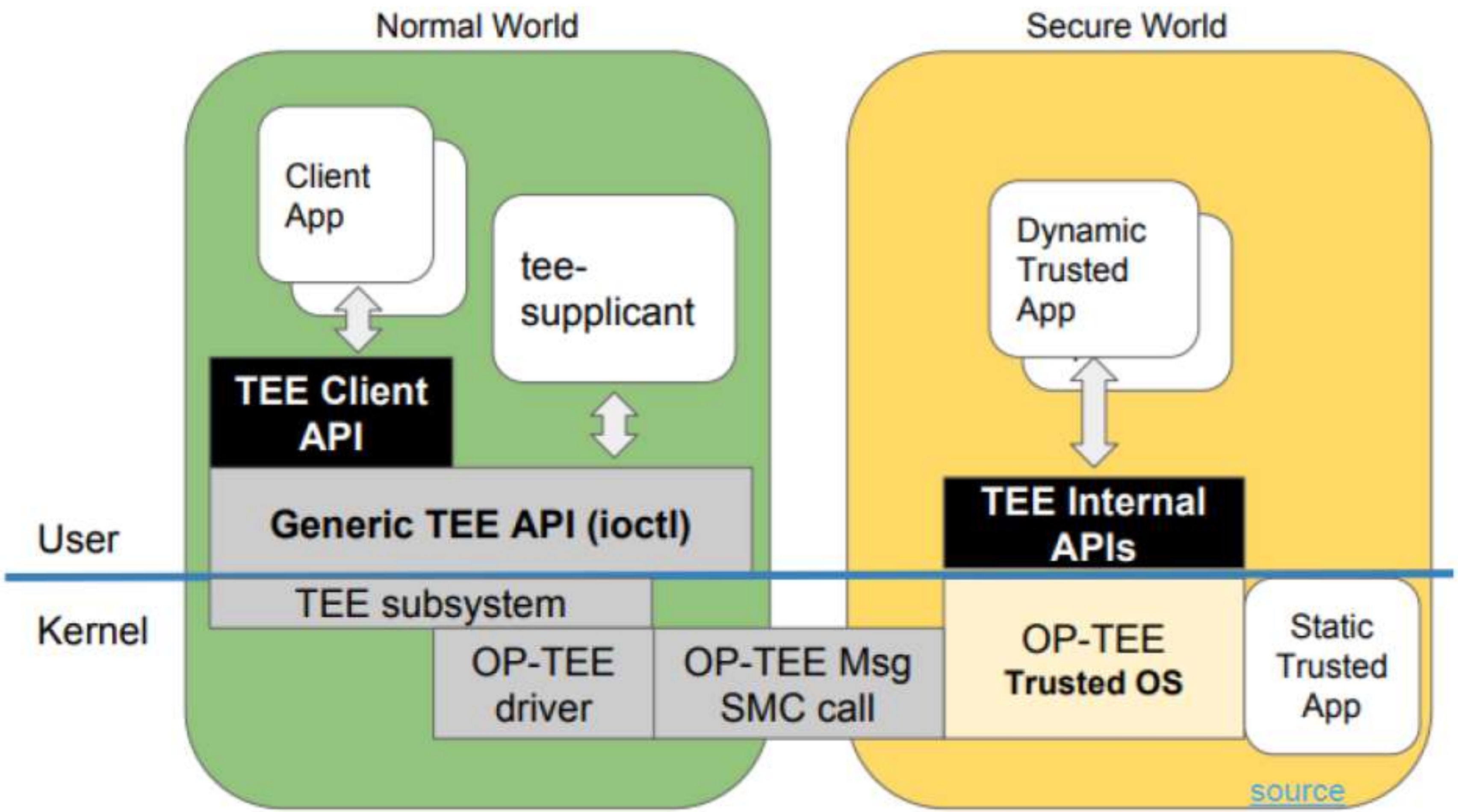


Cortex-A

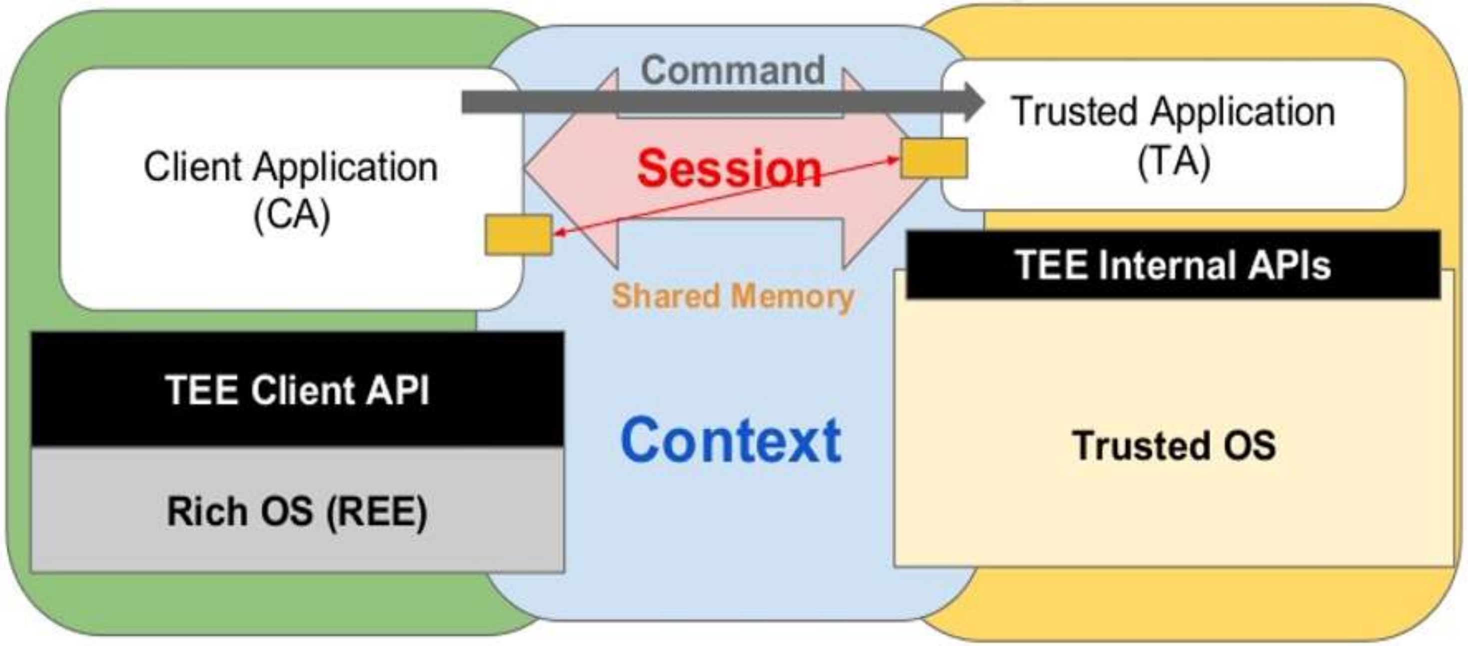


Cortex-M

OP-TEE Architecture



Communication



Context: Connection from CA to the Trusted OS.

Session: Connection from CA to TA.

Command: Unit of Communication from CA to TA.

Shared Memory: Shared buffers allocated by Client API or by CA and registered.

Booting Sequence

ROM Bootloader

Initializes critical peripherals like memory units, establishes initial trust

Secure World Boot

Boots from flash device, sets CP15SDISABLE register to lock configuration

Normal World Boot

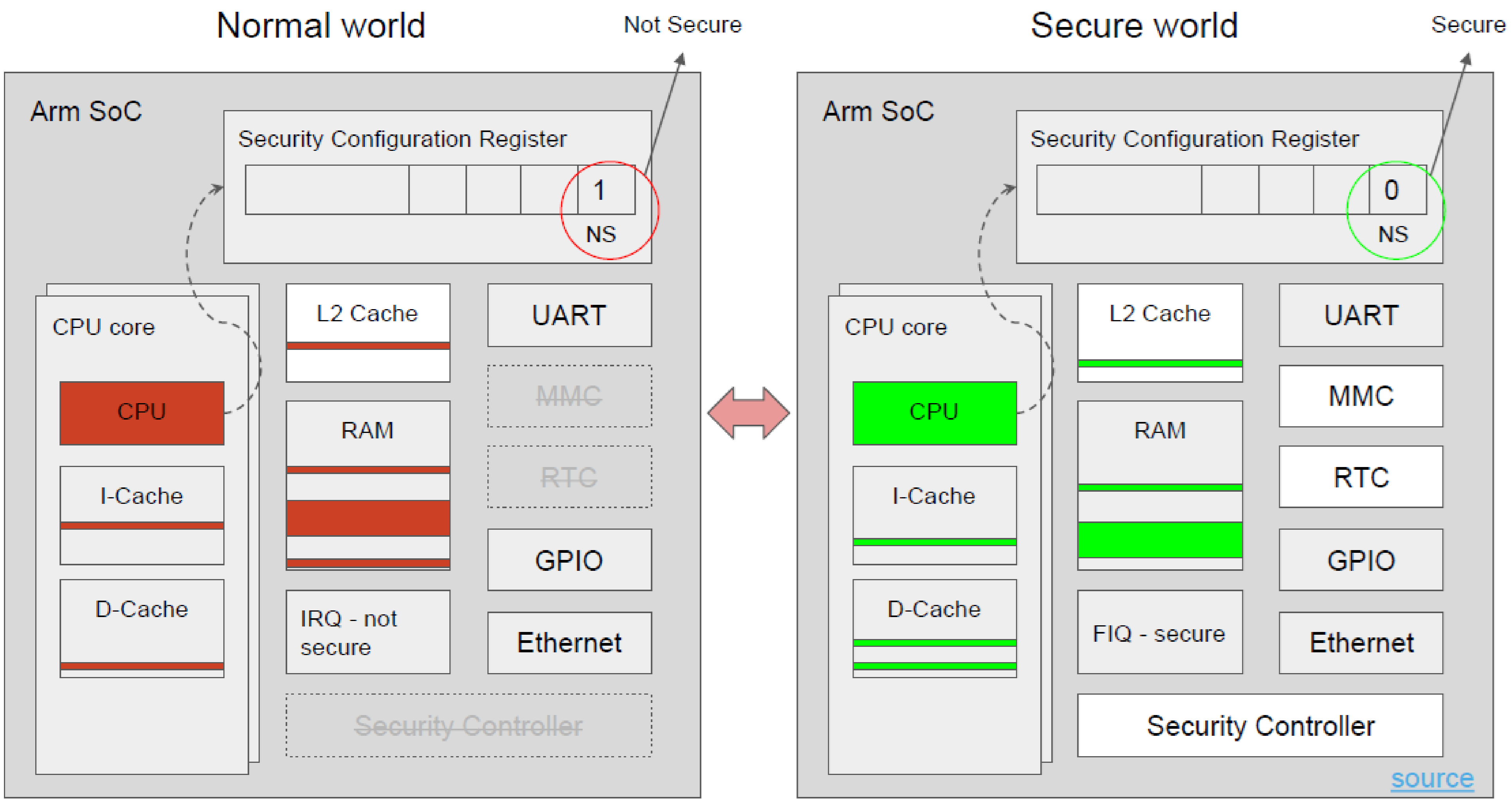
General purpose OS loads with restricted access to secure resources

Monitor Mode

Acts as context switch between worlds via SMC instruction, hardware interrupts, or external abort signals

- Each boot step is cryptographically verified using vendor-specific public keys, building a chain of trust.
- One-Time Programmable (OTP) techniques generate unique keys for each device component.

Hardware Support



Memory and Interrupt Management

- Memory Controllers
  - TZASC
    - TrustZone Address Space Controller allows dynamic classification of memory-mapped devices as secure or non-secure, enabling arbitrary partitions
  - TZMA
    - TrustZone Memory Adapter divides on-chip static memory into secure and non-secure halves in 4KB multiples (max 2MB total)
- Generic Interrupt Controller
  - The GIC handles secure and non-secure prioritized interrupts while preventing unauthorized access.
  - To prevent denial-of-service attacks, it only handles interrupts in the lower half of the priority hierarchy.